

LANDSCAPES OF THE CHURCH

05 — Sparse & Extreme Landscapes

Diocesan and Episcopal Conference layers: overview, ArcGIS Online-ready descriptions, methodology, QC, formulas, workflow, and reproducible code

LSAP Integral Ecology Atlas | Technical Documentation | 3 October 2026

10. Analysis 05 — Sparse & Extreme Landscapes

10.1 Diocesan layer overview

Layer name	LandscapesChurch_05_Dioceses_SparseExtreme_V2
Layer alias	05 - Diocesan Sparse & Extreme Landscapes
Scale / features	Diocese; 2,874 features
Description	Combines bare land and snow/ice to identify sparsely vegetated, arid, rocky, alpine, glaciated, and cryospheric landscape contexts without treating these systems as inherently degraded.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Bare + snow/ice combined extent; component shares; leading component; descriptive extent class; analytical rank/percentile fields.
Key story fields	STORY_SPARSE; STORY_CHARACTER; STORY_SCALE; STORY_CHURCH; STORY_SUMMARY; STORY_COMPREHENSIVE
Status	Publication-ready

10.2 Episcopal Conference layer overview

Layer name	LandscapesChurch_05_Conferences_SparseExtreme_V2
Layer alias	05 - Episcopal Conference Sparse & Extreme Landscapes
Scale / features	Episcopal Conference; 122 features
Description	Summarizes bare and snow/ice systems across Episcopal Conference geographies and interprets their physical landscape context without assigning ecological quality.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Bare + snow/ice extent; internal balance;

	Conference/Church/LSAP narrative context.
Key story fields	Diocesan story set plus STORY_LSAP.
Status	Publication-ready

10.3 Formulae and extent classes

Sparse/extreme share	$\text{SPARSE_PCT} = \text{PCT_LULC_Bare} + \text{PCT_LULC_SnowIce}$ Combined share of bare land and snow/ice.
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Bare share within sparse/extreme	$\text{BARE_SHARE_SPARSE} = 100 \times \text{Bare} / (\text{Bare} + \text{SnowIce})$ Internal component share.
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Snow/ice share within sparse/extreme	$\text{SNOW_SHARE_SPARSE} = 100 \times \text{SnowIce} / (\text{Bare} + \text{SnowIce})$ Internal component share.
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- <0.1% — Trace
- 0.1–<1% — Limited
- 1–<5% — Present
- 5–<25% — Substantial
- ≥25% — Extensive

The extent class is analytical only and was intentionally omitted from final story prose.

10.4 Core QC

Retained QC output

SPARSE & EXTREME LANDSCAPES – DISTRIBUTION QC

Dioceses:

Median SPARSE_PCT: 0.041%
 95th percentile: 4.000%
 Maximum: 86.859%
 Bare primary: 98.7%
 Snow/Ice primary: 1.2%

Episcopal Conferences:

Median SPARSE_PCT: 0.090%
 95th percentile: 34.654%
 Maximum: 71.948%
 Bare primary: 94.3%
 Snow/Ice primary: 5.7%

Final narrative QC:

- All rank / percentile language removed from stories.
- Rank fields retained analytically.
- Bare land and snow/ice are not described as ecological degradation.
- Sede vacante grammar patched in diocesan stories.
- Final lean QC returned 0 issues.

Relevant reproducible code

A.8 Analysis 05 — Sparse & Extreme Landscapes

```
bare = safe(row["PCT_LULC_Bare"])
snow = safe(row["PCT_LULC_SnowIce"])
```

```
sparse_pct = bare + snow
```

```

if sparse_pct > 0:
    bare_share_sparse = 100.0 * bare / sparse_pct
    snow_share_sparse = 100.0 * snow / sparse_pct
else:
    bare_share_sparse = 0.0
    snow_share_sparse = 0.0

sparse_primary = (
    "Bare"
    if bare >= snow
    else "Snow/Ice"
)

def sparse_class(v):
    if v < 0.1: return "Trace"
    if v < 1.0: return "Limited"
    if v < 5.0: return "Present"
    if v < 25.0: return "Substantial"
    return "Extensive"

```

A.10 Safe m² → km² conversion

```

import statistics

values = []

with arcpy.da.SearchCursor(
    feature_class,
    ["ANALYSIS_AREA"]
) as cursor:

    for (value,) in cursor:
        if value is not None and value > 0:
            values.append(float(value))

median_area = statistics.median(values)

if median_area >= 1_000_000:

    with arcpy.da.UpdateCursor(
        feature_class,
        ["ANALYSIS_AREA"]
    ) as cursor:

        for row in cursor:
            if row[0] is not None:
                row[0] = float(row[0]) / 1_000_000.0
                cursor.updateRow(row)

else:
    print("Area already appears to be in km²; no conversion performed.")

```

A.11 Rank / percentile reference implementation

```

def competition_rank(values):
    """
    values = iterable of (feature_id, numeric_value)
    Rank 1 = highest.
    Ties receive the same rank; the next rank skips accordingly.
    """
    ordered = sorted(
        values,
        key=lambda x: x[1],
        reverse=True
    )

    result = {}
    previous = None
    current_rank = 0

    for position, (feature_id, value) in enumerate(ordered, start=1):
        if previous is None or value != previous:

```

```

        current_rank = position

        result[feature_id] = current_rank
        previous = value

    return result

def percentile_at_or_below(all_values, value):
    valid = [v for v in all_values if v is not None]

    if not valid:
        return None

    n = sum(v <= value for v in valid)

    return 100.0 * n / len(valid)

```

A.16 Publication cleanup pattern

Keep only fields intentionally required for the publication derivative.

```

KEEP_FIELDS = {
    "BUFFER_ID",
    "Account_Name",
    "Country",
    "Language",
    "Official_website",
    "Sector",
    "BUFFER_AREA",
    "LANDSCAPE_SCALE",
    # ... eight PCT_LULC fields ...
    # ... coverage/mosaic/transition fields ...
    # ... story fields ...
}

delete_fields = [
    f.name
    for f in arcpy.ListFields(feature_class)
    if (
        not f.required and
        f.name not in KEEP_FIELDS
    )
]

if delete_fields:
    arcpy.management.DeleteField(
        feature_class,
        delete_fields
    )

# Always run a final required-field check after deletion.

```

Sources and attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas / project Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and the underlying credited source organizations.

Esri Global Biodiversity and Conservation Hexagons: <https://www.arcgis.com/home/item.html?id=f1bdbcac3c54ad8b5b21f5b17d91e8d>

Esri 2026 update: <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>

Summarize Within documentation: <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>